

PERFORMANCE CHARACTERISTICS TEST OF VARIABLE SPEED CENTRIFUGAL PUMP

AIM:-

To conduct the performance test on a variable speed centrifugal pump and draw the following graphs to obtain the main characteristic curves of a pump operating at different speeds.

1. Head v/s Discharge
2. Input Power v/s Discharge
3. Output Power v/s Discharge
4. Efficiency v/s Discharge
5. Efficiency v/s Speed

PRINCIPLE

For each speed, the rate of flow or the discharge is varied by means of delivery valve and for each value of the discharge the corresponding values of head, input power, output power and the efficiency are calculated. The procedure can be repeated for different speeds of the pump.

Total head, $H = H_d + H_s + l_d$

H_d = Delivery Pressure head in meters of water

H_s = Suction Pressure head in meters of water

l_d = Level difference between delivery and suction pressure heads in meters of water

Discharge, Q

$$Q = \frac{A R m^3}{t s}$$

Where A = Area of the collecting tank = $0.7 \times 0.4 \text{ m}^2$

R = Rise of water level in the collecting tank = 0.1 m

t = Time in seconds for R

$$\text{Overall efficiency of pump, } \eta = \frac{\text{Output Power}}{\text{Input Power}} \times 100 = \frac{OP}{IP} \times 100$$

$$\text{Input power, } IP = \frac{n 3600 \eta_M \eta_T}{T E_m} \text{ kW}$$

OBSERVATIONS

[illegible]

Where

- n = No of impulses of energy meter
 T = Time for 'n' impulses of energy meter, in s
 E_m = Energy meter constant impulse/kW hour
 η_M = Motor efficiency = 80%
 η_T = Transmission efficiency = 90%

Output power, $OP = \frac{\rho g Q H}{1000} \text{ kW}$

Where

- ρ = Density of flowing liquid in kg/m^3
 g = Acceleration due to gravity in m/s^2

Q = Discharge of flowing liquid in m^3/s

H = Total head in m = $H_d + H_s + l_d$

H_d = Delivery Pressure head in meters of water

H_s = Suction Pressure head in meters of water

l_d = Level difference between delivery and suction pressure heads in meters of water

PRECAUTIONS

- Never run the pump without liquid in it as this would cause damage to stuffing box, bush bearing etc.
- Never try to throttle the suction side of the pump to control the discharge as it would seriously affect the performance of the pump.

PROCEDURE

1. Before starting the pump, set the belt on required cone pulley.
2. The delivery valve, pressure gauge valve and vacuum gauge valve are fully closed.
3. Start the pump and open the delivery valve fully. Open the gauges valves to allow the flow to stabilise.
4. Note down the pressure gauge and vacuum gauge readings.
5. Note down the time for 'n' rise of water in the collecting tank.
6. Note down the time for 'n' impulse of the energy meter, and also note the speed of pump by using a tachometer.
7. The above procedure is repeated by gradually decreasing the flow of water by controlling the delivery valve (take minimum 6 sets of readings)
8. Now shift the belt to next groove of pulley and repeat the experiment.

SAMPLE CALCULATIONS (Set No :)

Discharge in m³/s

$$Q = \frac{AR}{t} \frac{m^3}{s}$$

Where A = Area of the collecting tank = 0.7 x 0.4 m²

R = Rise of water level in the collecting tank = 0.1 m

t = Time in seconds for R

$$Q = \frac{AR}{t} = \frac{m^3}{s}$$

Total head, H = H_d + H_s + l_d

H_d = Delivery Pressure head in meters of water

H_s = Suction Pressure head in meters of water

l_d = Level difference between delivery and suction pressure heads in meters of water

h_d - Delivery pressure head, (Kgf/cm²)

h_s - Suction pressure head, (mm of Hg)

ρ_w - Density of water, 1000 kg/m³

ρ_{Hg} - Density of Hg, 13600 kg/m³

Delivery Pressure head $H_d = \frac{h_d}{\rho_w \times 10^{-4}} =$

Suction Pressure head $H_s = \frac{h_s \times 10^{-3} \times \rho_{Hg}}{\rho_w} =$

l_d =

Total Head, H in meters = H_d + H_s + l_d =

➤ Out Put Power, OP in kW $OP = \frac{\rho g Q H}{1000} \text{ kW}$

Density of water, ρ = 1000 kg/m³

Acceleration due to gravity, g = 9.81 m/s²
 Discharge, Q , m³/s =
 Total head, H in m =

Output Power, OP = $\frac{\rho g Q H}{1000}$ = kW

➤ Input Power, IP = $\frac{n \cdot 3600 \cdot \eta_M \cdot \eta_T}{TE_m}$

No of impulses of Energy meter, n =
 Motor Efficiency, η_M = 80%
 Transmission efficiency, η_T = 90 %
 Time for 'n' impulse of Energy meter, T = sec
 Energy meter Constant, E_m = Imp / kW hour.

Input power, IP in kW = = kW

➤ Overall Efficiency, η_o = $\frac{OP}{IP} \times 100$ = %

GRAPHS

1. Head v/s Discharge
2. Input Power v/s Discharge
3. Output Power v/s Discharge
4. Efficiency v/s Discharge

RESULT:-

The required graphs are plotted.

INFERENCE:-